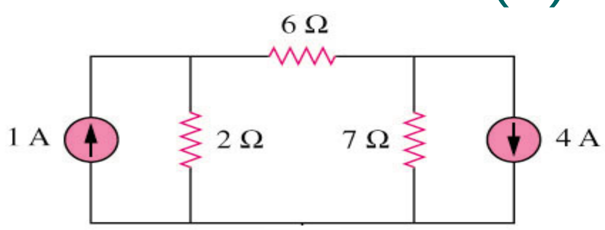
Methods of Analysis
Ch 3

3.1 Motivation (1)



If you are given the following circuit, how can we determine

1) the voltage across each resistor,
2) current through each resistor.
3) power generated by each current source, etc.

What are the things which we need to know in order to determine the answers?

3.1 Motivation (2)
The LAWS

Things we need to know in solving any resistive circuit with current and voltage sources only:

- Kirchhoff's Current Laws (KCL)
- Kirchhoff's Voltage Laws (KVL)
- Ohm's Law

How should we apply these laws to determine the answers?

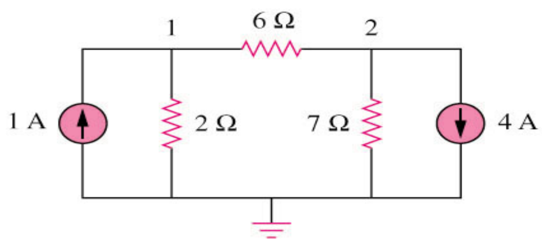
3.1 Motivation (3)
THESE TWO TECHNIQUES

KCL $\longrightarrow$ Nodal Analysis
aka (Voltage Node Analysis)

KVL $\longrightarrow$ Mesh (Current) Analysis

- Can be used in any linear circuit
- To obtain a set of simultaneous equations
- Equations solved using:
  - Substitution
  - Cramer's rule (Appendix A)

3.2 Nodal Analysis
Why?



- It provides a general procedure for analyzing circuits using node voltages as opposed to element voltages as the circuit variables.
- Reduces the number of equations required to solve a circuit.

3.2 Nodal Analysis
STEPS

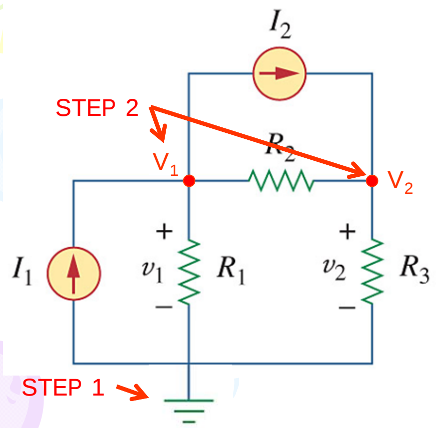
Steps to determine the node voltages:

1. Select a node as the reference node *(ground)*.
2. Assign voltages $V_1, V_2, \ldots, V_{n-1}$ to the remaining “critical” nodes.
   - “Critical node” = node with more than 2 branches attached.
   - The voltages are referenced with respect to the reference node. I.e. Measured between Node and Ground.
3. Apply KCL to each of the non-reference nodes. Use Ohm’s law to express the branch currents in terms of node voltages.
4. Solve the resulting simultaneous equations to obtain the unknown node voltages.

3.2 Nodal Analysis: EXAMPLE 1

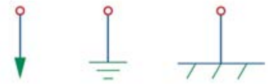
Calculate v_1 & v_2

Two unknown: v_1, v_2; requires two algebraic expression to calculate
Using Nodal voltage analysis we will obtain these two algebraic expressions



STEP 1:
Select a Reference Node
WHERE??

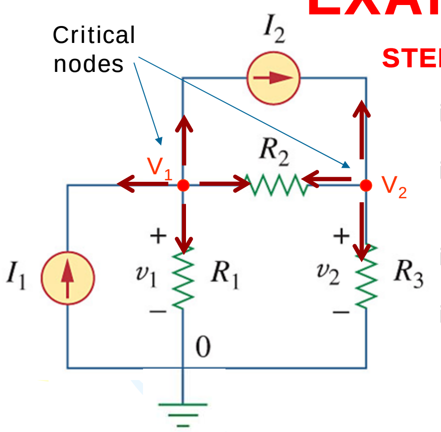
- Negative terminal of V-source
- Entry terminal of I-source
- Node with many branches
- Bottom node (many cases)



STEP 2:
Name critical nodes: V_1, V_2, …, V_n

3.2 Nodal Analysis

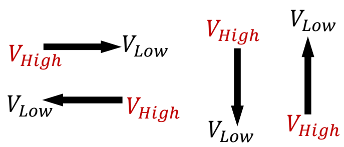
EXAMPLE 1



STEP 3: Apply KCL to each node

i. Assign a current to each branch at each critical node.
ii. All currents to point **OUT** of the node.
iii. Remember: Current flows from HIGH to LOW voltage ∴
iv. All branch currents to be written ito Ohm's law: $i = \frac{V_{High} - V_{Low}}{R_{Branch}}$

i. Number of critical nodes = number of equations.
ii. Number of branches = number of terms.

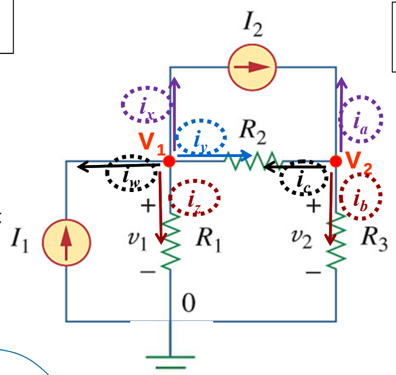


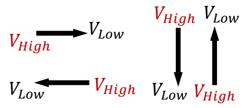
EXAMPLE 1

STEP 3: Apply KCL to each node

$$i = \frac{V_{High} - V_{Low}}{R_{Branch}}$$

$$i_{branch} = \frac{V_{n\_from} - V_{n\_to}}{R_{branch}}$$





KCL@ Node 1 (4 Branches):

$i_w + i_x + i_y + i_z$

Replace the branch current using Ohm's law

$i_w = -I_1$;

$i_x = I_2$;

$i_y = \frac{V1 - V2}{R2}$; $i_z = \frac{V1}{R1}$

$$(-I_1) + (I_2) + \frac{V_1 - V_2}{R_2} + \frac{V_1 - 0}{R_1} = 0$$

Negative because outgoing branch current is opposite to the source's current

KCL@ Node 2 (3 Branches):

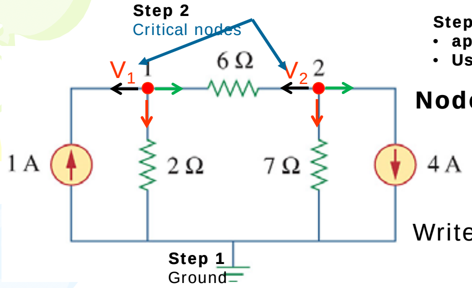
$i_a + i_b + i_c = 0$

Replace the branch current using Ohm's law

$$(-I_2) + \frac{V_2 - 0}{R_3} + \frac{V_2 - V_1}{R_2} = 0$$

3.2 Nodal Analysis

EXAMPLE 2



Step 3:
- **apply KCL @ each node**
- **Use Ohm's law and replace branch current**

Node 1: $i_{1A} + i_{2\Omega} + i_{6\Omega} = 0$

$$\text{X6}\{-1 + \frac{V_1 - 0}{2} + \frac{V_1 - V_2}{6} = 0$$

$$-6 + 3V_1 + V_1 - V_2 = 0$$

Write ito: $a_1V_1 + b_1V_2 = k_1$

$$4V_1 + (-1)V_2 = 6 \quad (1)$$

Node 2: $i_{6\Omega} + i_{7\Omega} + i_{4A} = 0$

$$\text{X42}\{\frac{V_2 - V_1}{6} + \frac{V_2 - 0}{7} + 4 = 0$$

$$7V_2 - 7V_1 + 6V_2 + 168 = 0$$

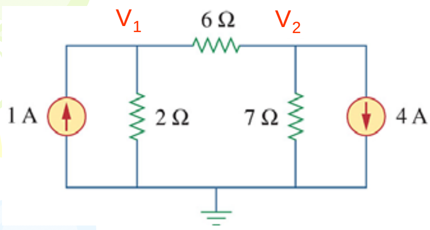
Write ito: $a_2V_1 + b_2V_2 = k_2$

$$(-7)V_1 + 13V_2 = -168 \quad (2)$$

$$4V_1 + (-1)V_2 = 6 \quad (1)$$
$$(-7)V_1 + 13V_2 = -168 \quad (2)$$

$a_1 = 4$; $b_1 = -1$
$a_2 = -7$; $b_2 = 13$

Ex. 2 – Method 1: Elimination



$$4V_1 + (-1)V_2 = 6 \quad (1)$$
$$(-7)V_1 + 13V_2 = -168 \quad (2)$$

From (1): $V_2 = 4V_1 - 6$ (3)

(3) in (2): $(-7)V_1 + 13(4V_1 - 6) = -168$

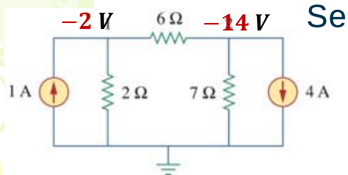
$$45V_1 - 78 = -168$$
$$45V_1 = -90 \qquad \boldsymbol{V_1 = -2\,V}$$

From (3): $V_2 = 4(-2) - 6$

$$\boldsymbol{V_2 = -14\,V}$$

Ex. 2 – Method 2: Cramer's Rule

See Appendix A



$$4\ V_1 + -1\ V_2 = 6 \quad (1)$$
$$-7\ V_1 + 13\ V_2 = -168 \quad (2)$$

$$\begin{bmatrix} 4 & -1 \\ -7 & 13 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -168 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 4 & -1 \\ -7 & 13 \end{vmatrix}$$

$(-1 \times -7) = 7$ $\qquad (4 \times 13) = 52$

$\Delta = 52 - 7 \qquad \Delta = 45$

$$V_1 = \frac{\Delta_1}{\Delta} \qquad \Delta_1 = \begin{vmatrix} 6 & -1 \\ -168 & 13 \end{vmatrix}$$

(-1×-168) $= 168$ $\qquad (6 \times 13)$ $= 78$

$\Delta_1 = 78 - 168 \qquad \Delta_1 = -90$

$$\boxed{V_1 = \frac{-90}{45} = -2\,V}$$

$$V_2 = \frac{\Delta_2}{\Delta} \qquad \Delta_2 = \begin{vmatrix} 4 & 6 \\ -7 & -168 \end{vmatrix}$$

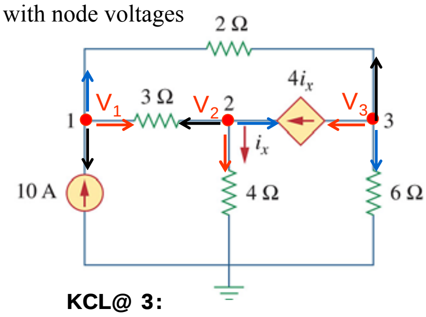
(6×-7) $= -42$ $\qquad (4 \times -168)$ $= -672$

$\Delta_2 = -672 - (-42) \qquad \Delta_2 = -630$

$$\boxed{V_2 = \frac{-630}{45} = -14\,V}$$

PP 3-2: Determine the node voltages.

Three unknown: V1, V2 & V3; requires three algebraic expression
Apply KCL @ each node
Using Ohm's law replace the branch currents with node voltages



KCL@ node 1:

$$\boldsymbol{i_{10A} + i_{3\Omega} + i_{2\Omega} = 0;}$$

Ohm's law: replace current with voltages

$$\text{x6}\{-\mathbf{10} + \frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{2} = 0$$

$$-60 + 2V_1 - 2V_2 + 3V_1 - 3V_3 = 0$$

$$5V_1 - 2V_2 - 3V_3 = 60 \quad (1)$$

KCL@ 2:

$$\boldsymbol{i_{3\Omega} + i_{4\Omega} + id = 0}$$

Ohm's law: replace current with voltages

$$\text{x12}\{\frac{V_2 - V_1}{3} + \frac{V_2 - 0}{4} - \mathbf{4}\boldsymbol{i_x} = 0$$

$$4V_2 - 4V_1 + 3V_2 - 48\frac{\boldsymbol{V_2}}{\mathbf{4}} = 0$$

$$-4V_1 - 5V_2 + 0V_3 = 0 \quad (2)$$

KCL@ 3:

$$\boldsymbol{i_d + i_{2\Omega} + i_{6\Omega} = 0}$$

Ohm's law: replace current with voltages

$$\text{x6}\{\ 4\boldsymbol{i_x} + \frac{V_3 - V_1}{2} + \frac{V_3 - 0}{6} = 0$$

$$24\frac{\boldsymbol{V_2}}{\mathbf{4}} + 3V_3 - 3V_1 + V_3 = 0$$

$$-3V_1 + 6V_2 + 4V_3 = 0 \quad (3)$$

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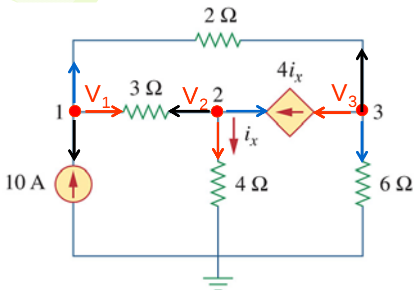
PP 3-2: Determine the node voltages.



$$5V_1 - 2V_2 - 3V_3 = 60 \quad (1)$$

$$-4V_1 - 5V_2 + 0V_3 = 0 \quad (2)$$

$$-3V_1 + 6V_2 + 4V_3 = 0 \quad (3)$$

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta} \quad V_2 = \frac{\Delta_2}{\Delta} \quad V_3 = \frac{\Delta_3}{\Delta}$$

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PP 3-2 … Calculating Δ

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \mathbf{60} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\Delta = \begin{vmatrix} +5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & +6 & +4 \end{vmatrix} \qquad \begin{matrix} +5 & -2 & -3 \\ -4 & -5 & 0 \end{matrix}$$

$(-3 \cdot -5 \cdot -3) = -45$

$(0 \cdot 6 \cdot 5) = 0$

$(4 \cdot -2 \cdot -4) = -32$

$+\ (5 \cdot -5 \cdot 4) = -100$

$+\ (-4 \cdot 6 \cdot -3) = 72$

$+\ (-3 \cdot -2 \cdot 0) = 0$

$$\Delta = (-100 + 72 + 0) - (-45 + 0 + 32)$$
$$= (-28) - (-13)$$
$$\Delta = -15$$

PP 3-2 … Calculating Δ_1

$\Delta = -15$

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \mathbf{60} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\Delta 1 = \begin{vmatrix} 60 & -2 & -3 \\ 0 & -5 & 0 \\ 0 & +6 & +4 \end{vmatrix} \qquad \begin{matrix} 60 & -2 & -3 \\ 0 & -5 & 0 \end{matrix}$$

$(-3 \cdot -5 \cdot 0) = 0$

$(0 \cdot 6 \cdot 0) = 0$

$(4 \cdot -2 \cdot 0) = 0$

$+\ (60 \cdot -5 \cdot 4) = -1200$

$+\ (0 \cdot 6 \cdot -3) = 0$

$+\ (0 \cdot -2 \cdot 0) = 0$

$$\Delta_1 = (-1200 + 0 + 0) - (0 + 0 + 0)$$
$$\Delta_1 = -1200$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-1200}{-15} = 80 \text{ V}$$

PP 3-2 … Calculating Δ_2

$\Delta = -15$

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \mathbf{60} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\Delta 2 = \begin{vmatrix} 5 & 60 & -3 \\ -4 & 0 & 0 \\ -3 & 0 & +4 \end{vmatrix}$$

$$\begin{array}{ccc} 5 & 60 & -3 \\ -4 & 0 & 0 \end{array}$$

$(-3 \cdot 0 \cdot 0) = 0$ $\quad + \ (5 \cdot 0 \cdot 4) = 0$

$(0 \cdot 0 \cdot 5) = 0$ $\quad + \ (-4 \cdot 0 \cdot -3) = 0$

$(4 \cdot 60 \cdot -4) = -960$ $\quad + \ (-3 \cdot 60 \cdot 0) = 0$

$$\Delta 2 = (0 + 0 + 0) - (0 + 0 - 960)$$

$$\Delta 2 = -960$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{-960}{-15} = 64 \text{ V}$$

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PP 3-2 … Calculating Δ_3

$\Delta = -15$

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} \mathbf{60} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\Delta 3 = \begin{vmatrix} 5 & -2 & 60 \\ -4 & -5 & 0 \\ -3 & 6 & 0 \end{vmatrix}$$

$$\begin{array}{ccc} 5 & -2 & 60 \\ -4 & -5 & 0 \end{array}$$

$(60 \cdot -5 \cdot -3) = 900$ $\quad + \ (5 \cdot -5 \cdot 0) = 0$

$(0 \cdot 6 \cdot 5) = 0$ $\quad + \ (-4 \cdot 6 \cdot 60) = -1440$

$(0 \cdot -2 \cdot -4) = 0$ $\quad + \ (-3 \cdot -2 \cdot 0) = 0$

$$\Delta 3 = (0 - 1440 + 0) - (900 + 0 + 0)$$

$$\Delta 3 = -2340$$

$$V_3 = \frac{\Delta_3}{\Delta} = \frac{-2340}{-15} = 156 \text{ V}$$

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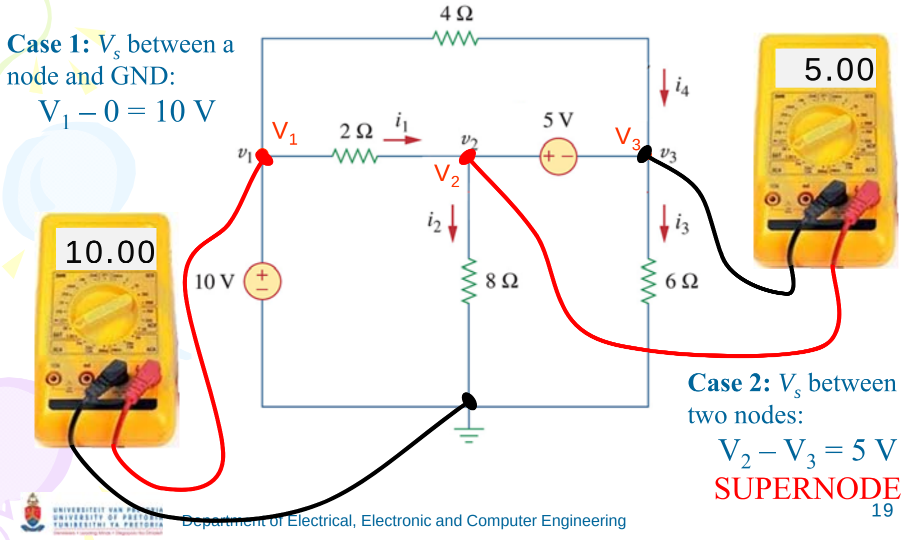
L#2: 3.3 Nodal Analysis With Voltage sources

Case 1: V_s between a node and GND:

$V_1 - 0 = 10$ V



Case 2: V_s between two nodes:

$V_2 - V_3 = 5$ V

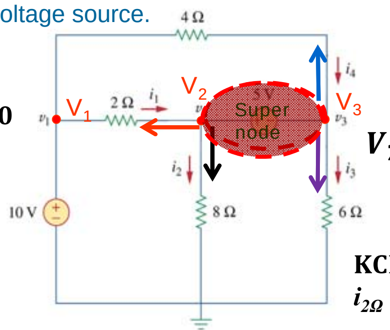
SUPERNODE

3.3 Nodal Analysis with Voltage sources

SUPERNODE

Formed by enclosing a (dependent or independent) **voltage** source connected between two non-reference nodes along with any elements connected in **parallel** with that voltage source.

$V_1 = 10$



$V_2 - V_3 = 5$ (1)

Note:

$i_1 = -\,i_{2\Omega} = \frac{V1-V2}{2}$

$i_2 = +\,i_{8\Omega} = \frac{V2}{8}$

$i_3 = +\,i_{6\Omega} = \frac{V3}{6}$

$i_4 = -\,i_{4\Omega} = \frac{V1-V3}{4}$

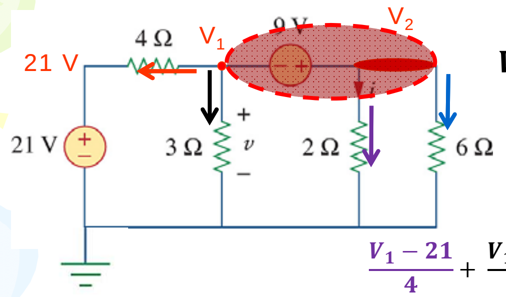
KCL at SUPERNODE:

$i_{2\Omega} + i_{8\Omega} + i_{6\Omega} + i_{4\Omega} = 0$

$$\frac{V_2 - 10}{2} + \frac{V_2 - 0}{8} + \frac{V_3 - 0}{6} + \frac{V_3 - 10}{4} = 0$$

Solve for V_2 and V_3 from (1) and (2) (2)

PP 3-3: Find v and i in the circuit.



$$V_2 - V_1 = 9 \quad (1)$$

KCL at SUPERNODE:

$$i_{4\Omega} + i_{3\Omega} + i_{2\Omega} + i_{6\Omega} = 0$$

$$\frac{V_1 - 21}{4} + \frac{V_1 - 0}{3} + \frac{V_2 - 0}{2} + \frac{V_2 - 0}{6} = 0 \quad (2)$$

Solve for V_2 and V_3 from (1) and (2)

$$-V_1 + V_2 = 9 \quad (1)$$
$$7V_1 + 8V_2 = 63 \quad (2)$$

$$x7\{ \quad -7V_1 + 7V_2 = 63$$
$$7V_1 + 8V_2 = 63$$
$$15V_2 = 126$$
$$V_2 = 8.4\ V$$

$$V_1 = V2 - 9$$
$$= 8.4 - 9$$
$$\therefore V_1 = v = -0.6\ V$$

$$i = \frac{8.4}{2} = 4.2\ A$$

Cramer's rule

$$\begin{bmatrix} -1 & 1 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 63 \end{bmatrix}$$

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PP 3-3: …

$$\begin{bmatrix} -1 & 1 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 63 \end{bmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta}$$

$$\Delta_1 = \begin{vmatrix} 9 & 1 \\ 63 & 8 \end{vmatrix}$$

$(1 \times 63) = 63$ $\quad$ $(9 \times 8) = 72$

$$\Delta_1 = 72 - 63 \qquad \Delta_1 = 9$$

$$\boxed{V_1 = \frac{9}{-15} = -0.6\ V}$$

$$\Delta = \begin{vmatrix} -1 & 1 \\ 7 & 8 \end{vmatrix}$$

$(1 \times 7) = 7$ $\quad$ $(-1 \times 8) = -8$

$$\Delta = -8 - 7 \qquad \Delta = -15$$

$$V_2 = \frac{\Delta_2}{\Delta}$$

$$\Delta_2 = \begin{vmatrix} -1 & 9 \\ 7 & 63 \end{vmatrix}$$

$(9 \times 7) = 63$ $\quad$ $(-1 \times 63) = -63$

$$\Delta_2 = -63 - (63) \qquad \Delta_2 = -126$$

$$\boxed{V_2 = \frac{-126}{-15} = 8.4\ V}$$

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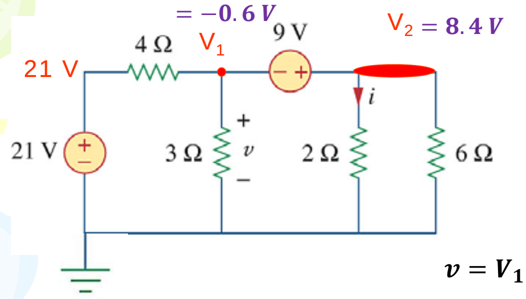
PP 3-3: Find v and i



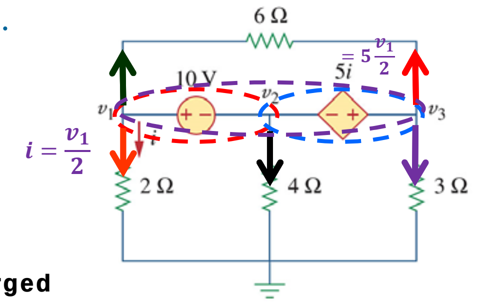
$$v = V_1 = -0.6\ V$$

$$\begin{bmatrix} -1 & 1 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 63 \end{bmatrix}$$

$$i = \frac{V_2}{2} = \frac{8.4}{2} = 4.2\ A$$

$V_1 = -0.6\ V$ $\quad V_2 = 8.4\ V$

PP 3-4 Find v_1, v_2, v_3 and i.



$$v_1 - v_2 = 10 \quad (1)$$

$$v_3 - v_2 = 5i$$

$$v_3 - v_2 = 5\frac{v_1}{2}$$

$$-5v_1 - 2v_2 + 2v_3 = 0 \quad (2)$$

Two supernode can be merged

KCL@SN

$$i_6 + i_2 + i_4 + i_3 + i_6 = 0$$

$$\cancel{\frac{v_1 - v_3}{6}} + \frac{v_1}{2} + \frac{v_2}{4} + \frac{v_3}{3} + \cancel{\frac{v_3 - v_1}{6}} = 0 \quad \}\text{x12}$$

$$6v_1 + 3v_2 + 4v_3 = 0 \quad (3)$$

$$\begin{bmatrix} 1 & -1 & 0 \\ -5 & -2 & 2 \\ 6 & 3 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

PP 3-4 … Calculating Δ

$$\begin{bmatrix} 1 & -1 & 0 \\ -5 & -2 & 2 \\ 6 & 3 & 4 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 1 & -1 & 0 \\ -5 & -2 & 2 \\ 6 & 3 & 4 \end{vmatrix}$$

$$\begin{matrix} 1 & -1 & 0 \\ -5 & -2 & 2 \end{matrix}$$

(0)

$(2 \cdot 3 \cdot 1) = 6$

$(4 \cdot -1 \cdot -5) = 20$

$(1 \cdot -2 \cdot 4) = -8$

$(-5 \cdot 3 \cdot 0) = 0$

$(6 \cdot -1 \cdot 2) = -12$

$$\Delta = (-8 - 12) - (6 + 20)$$
$$= (-20) - (26)$$
$$\Delta = -46$$

PP 3-4 … Calculating v_1 using Δ_1

$$\begin{bmatrix} 1 & -1 & 0 \\ -5 & -2 & 2 \\ 6 & 3 & 4 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} \qquad \Delta = -46$$

$$\Delta 1 = \begin{vmatrix} 10 & -1 & 0 \\ 0 & -2 & 2 \\ 0 & 3 & 4 \end{vmatrix}$$

$$\begin{matrix} 10 & -1 & 0 \\ 0 & -2 & 2 \end{matrix}$$

$(0) = 0$

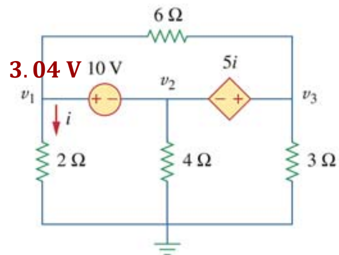
$(2 \cdot 3 \cdot 10)$
$= 60$

$(4 \cdot -1 \cdot 0)$
$= 0$

$(10 \cdot -2 \cdot 4)$
$= -80$

$(0) = 0$

$(0 \cdot) = 0$



$$\Delta_1 = (-80) - (60)$$
$$\Delta_1 = -140$$

$$i = \frac{3.04}{2} = 1.522\ A$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-140}{-46} = 3.044\ \text{V}$$

PP 3-2 … Calculating v_2 using Δ_2

$\Delta = -46$

$v_1 = 3.04$ V

$i = 1.52\ A$

$$\begin{bmatrix} 1 & -1 & 0 \\ -5 & -2 & 2 \\ 6 & 3 & 4 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta 2 = \begin{vmatrix} 1 & 10 & 0 \\ -5 & 0 & 2 \\ 6 & 0 & 4 \end{vmatrix}$$

$$\begin{matrix} 1 & 10 & 0 \\ -5 & 0 & 2 \end{matrix}$$

$(0) = 0$

$(0) = 0$

$(4 \cdot 10 \cdot -5)$
$= -200$

(0)

(0)

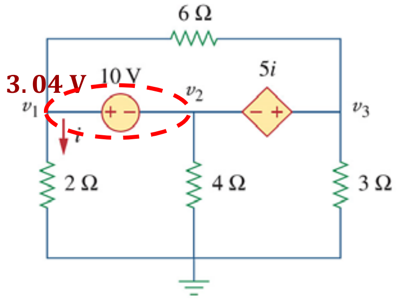
$(6.10.2)$
$= 120$

$\Delta 2 = (120) - (-200)$

$\Delta 2 = +320$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{+320}{-46} = -6.96\ \text{V}$$

Method#2 for calculating v_2



$v_1 - v_2 = 10$

$v_2 = v_1 - 10$

$= 3.04 - 10$

$v_2 = -6.96\ V$

PP 3-2 … Calculating v_3 using Δ_3

$\Delta = -46$

$$\begin{bmatrix} 1 & -1 & 0 \\ -5 & -2 & 2 \\ 6 & 3 & 4 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta 3 = \begin{vmatrix} 1 & -1 & 10 \\ -5 & -2 & 0 \\ 6 & 3 & 0 \end{vmatrix}$$

$$\begin{matrix} 1 & -1 & 10 \\ -5 & -2 & 0 \end{matrix}$$

$\left(\begin{matrix} 10 . -2 . 6 \\ = -120 \end{matrix}\right)$

(0)

(0)

(0)

$\left(\begin{matrix} -5.3.10 \\ = -150 \end{matrix}\right)$

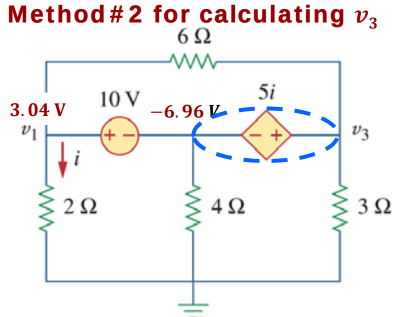
(0)

$\Delta 3 = (-150) - (-120)$

$\Delta 3 = -30$

$$V_3 = \frac{\Delta_3}{\Delta} = \frac{-30}{-46} = 0.65\ \text{V}$$

Method#2 for calculating v_3



$v_3 = 5i + v_2$

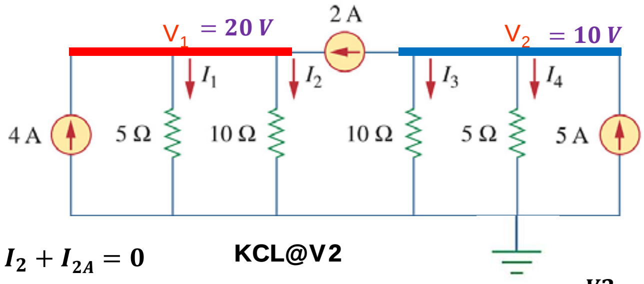
$i = \frac{3.044}{2} = 1.522\ A$

$v_3 = 5(1.522) - 6.96$

$v_3 = 0.65\ V$

Problem 3-4 Find I_1 through I_4.

$$I_1 = \frac{V1}{5}$$



KCL@V1

$$I_{4A} - I_1 - I_2 + I_{2A} = 0$$

$$4 - \frac{V_1}{5} - \frac{V_1}{10} + 2 = 0$$

$$V_1 = 20\,V$$

$$I_1 = \frac{20}{5} = 4\,A$$

$$I_2 = \frac{20}{10} = 2\,A$$

KCL@V2

$$-2 - I_3 - I_4 + 5 = 0$$

$$I_3 = \frac{V2}{5}$$

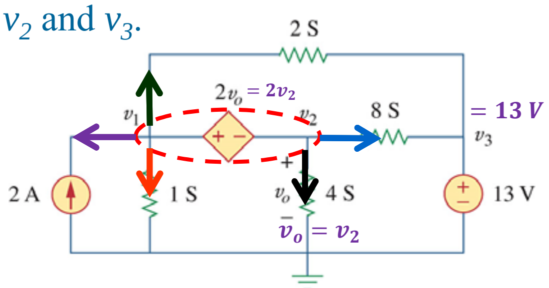
$$-\frac{V_2}{10} - \frac{V_2}{5} + 3 = 0$$

$$V_2 = 10\,V$$

$$I_3 = \frac{10}{10} = 1\,A$$

$$I_4 = \frac{10}{5} = 2\,A$$

Problem 3-16 Find v_1, v_2 and v_3.



$$v_1 - v_2 = 2v_o$$

$$v_1 - v_2 = 2(v_2)$$

$$v_1 = 3v_2 \quad (1)$$

KCL@SN

$$i_2 + i_{2A} + i_1 + i_4 + i_8 = 0$$

$$(v_1 - 13)2 - 2 + (v_1)1 + (v_2)4 + (v_2 - 13)8 = 0$$

$$3v_1 + 12v_2 = 132 \quad (2)$$

Substitute $v_1 = 3v_2$ *into* (2)

$$9v_2 + 12v_2 = 132$$

$$\rightarrow v_2 = 6.29\,V$$

$$v_1 = 3 * 6.29 = 18.87\,V$$

Class Test #1

Date:	19 February		
Time:	7:30		
Scope:	Ch01 & Ch02		
Venue:	Eng 3-1	Surnames:	A – K
	Eng 3-6	Surnames:	L – O
	IT 2-23	Surnames:	P – Z

L#3: 3.4 Mesh Analysis

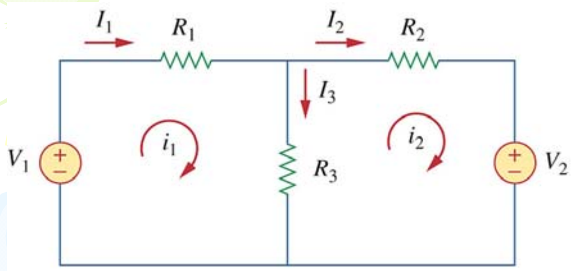
Why?

1. Mesh analysis provides another general procedure for analyzing circuits using mesh currents as the circuit variables.

2. Nodal analysis applies KCL to find unknown voltages in a given circuit, while mesh analysis applies KVL to find unknown currents.

3. A mesh is a loop which does not contain any other loops within it.

3.4 Mesh Analysis
STEPS

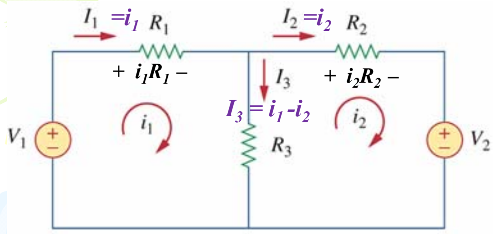


$I_1 = i_1$
$I_2 = i_2$
$I_3 = i_1 - i_2$

Steps to determine the mesh currents:

1. Assign mesh currents $i_1, i_2, \ldots,$ in to the n meshes – clock wise.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

3.4 Mesh Analysis
STEPS: KVL in Loops



Mesh current i_1 & i_2
Branch Current: I_1, I_2 & I_3
$P_{V1} = -i_1 V_1$
$I_1 = i_1, I_2 = i_2$ & $I_3 = i_1 - I_2$
$V_{R3} = R_3 I_3$
How to calculate i_1 & i_2 **?**

R_3 has 2 currents flowing through it: i_1 and i_2

KVL@Loop 1:

$-V_1 + V_{R1} + V_{R3} = 0$
$-V_1 + R_1 i_1 + R_3 \boxed{(i_1 - i_2)} = 0$

I_{net} through R_3 wrt loop 1

Write ito:

$a_1 i_1 + b_1 i_2 = k_1$
$(R_1 + R_3) i_1 + (-R_3)\, i_2 = V_1$

$R(i_{loop} - i_{opposing\_loop})$

KVL@Loop 2:

$V_2 + V_{R3} + V_{R2} = 0$
$V_2 + R_3 \boxed{(i_2 - i_1)} + R_2 i_2 = 0$

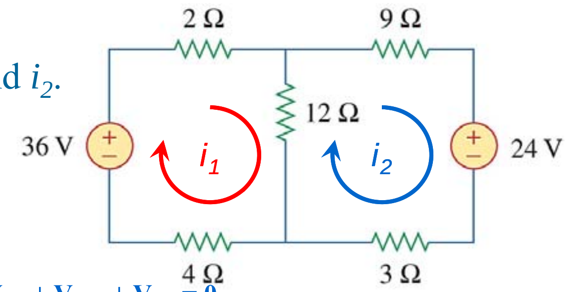
I_{net} through R_3 wrt loop 2

Write ito:

$a_2 i_1 + b_2 i_2 = k_2$
$(-R_3) i_1 + (R_2 + R_3)\, i_2 = -V_2$

PP 3-5

Determine i_1 and i_2.



KVL@Loop 1:

$$V_{36V} + V_{2\Omega} + V_{12\Omega} + V_{4\Omega} = 0$$
$$-36 + 2i_1 + 12(i_1 - i_2) + 4i_1 = 0$$
$$(2+12+4)i_1 + (-12)i_2 = 36$$
$$18i_1 - 12i_2 = 36 \qquad (1)$$

KVL@Loop 2:

$$V_{24V} + V_{3\Omega} + V_{12\Omega} + V_{9\Omega} = 0$$
$$24 + 3i_2 + 12(i_2 - i_1) + 9i_2 = 0$$
$$-12i_1 + (3+12+9)i_2 = -24$$
$$-12i_1 + 24i_2 = -24 \qquad (2)$$

Matrix: $\begin{bmatrix} 18 & -12 \\ -12 & 24 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 36 \\ -24 \end{bmatrix}$

$$\Delta = \begin{vmatrix} 18 & -12 \\ -12 & 24 \end{vmatrix} = 288$$

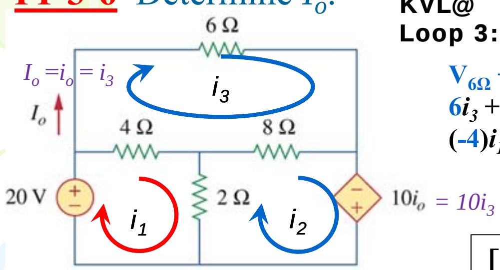
$$\Delta_1 = \begin{vmatrix} 36 & -12 \\ -24 & 24 \end{vmatrix} = 576$$

$$\boxed{i_1 = \frac{\Delta_1}{\Delta} = \frac{576}{288} = 2\ A}$$

$$\Delta_2 = \begin{vmatrix} 18 & 36 \\ -12 & -24 \end{vmatrix} = 0$$

$$\boxed{i_2 = \frac{\Delta_2}{\Delta} = \frac{0}{288} = 0\ A}$$

PP 3-6 Determine I_o.



KVL@ Loop 3:

$$V_{6\Omega} + V_{8\Omega} + V_{4\Omega} = 0$$
$$6i_3 + 8(i_3 - i_2) + 4(i_3 - i_1) = 0$$
$$(-4)i_1 + (-8)i_2 + (18)i_3 = 0 \qquad (3)$$

$$\boxed{\begin{bmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}}$$

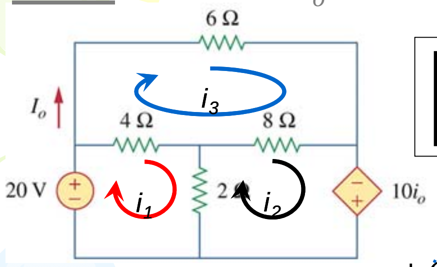
KVL@ Loop 1:

$$V_{20V} + V_{4\Omega} + V_{2\Omega} = 0$$
$$-20 + 4(i_1 - i_3) + 2(i_1 - i_2) = 0$$
$$(4+2)i_1 + (-2)i_2 + (-4)i_3 = 20$$
$$6i_1 + (-2)i_2 + (-4)i_3 = 20 \qquad (1)$$

KVL@ Loop 2:

$$V_{10io} + V_{2\Omega} + V_{8\Omega} = 0$$
$$-10i_o + 2(i_2 - i_1) + 8(i_2 - i_3) = 0$$
$$-10i_3 + 2(i_2 - i_1) + 8(i_2 - i_3) = 0$$
$$(-2)i_1 + (10)i_2 + (-18)i_3 = 0 \qquad (2)$$

PP 3-6 Determine I_o cont. … calculate Δ



$$\begin{bmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{vmatrix}$$

$$\begin{matrix} 6 & -2 & -4 \\ -2 & 10 & -18 \end{matrix}$$

$(-4 \cdot 10 \cdot -4) = 160$

$(-18 \cdot -8 \cdot 6) = 864$

$(18 \cdot -2 \cdot -2) = 72$

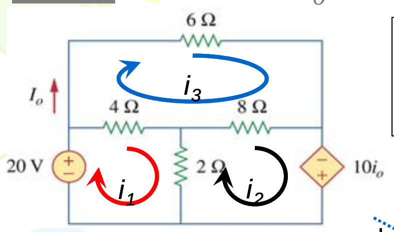
$(6 \cdot 10 \cdot 18) = 1080$

$(-2 \cdot -8 \cdot -4) = -64$

$(-4 \cdot -2 \cdot -18) = -144$

$$\Delta = (1080 - 64 - 144) - (160 + 864 + 72)$$
$$= \mathbf{-224}$$

PP 3-6 Determine I_o cont. … calculate $\Delta 3$



$$\begin{bmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}$$

Only I_o is asked so calculate Δ_3

$\Delta = -224$

$$\Delta_3 = \begin{vmatrix} 6 & -2 & 20 \\ -2 & 10 & 0 \\ -4 & -8 & 0 \end{vmatrix}$$

$$\begin{matrix} 6 & -2 & 20 \\ -2 & 10 & 0 \end{matrix}$$

$(20 \cdot 10 \cdot -4) = -800$

(0)

(0)

(0)

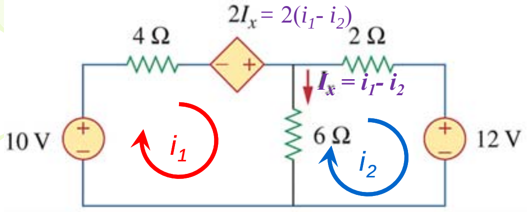
$(-2 \cdot -8 \cdot 20) = 320$

(0)

$$\Delta_3 = (320) - (-800)$$
$$= \mathbf{1120}$$

$$I_o = i_3 = \frac{\Delta_3}{\Delta} = \frac{1120}{-224} = \mathbf{-5}\ A$$

Problem 3-39 Determine i_1 and i_2 (Mesh Analysis)



Loop 1:

$V_{10V} + V_{4\Omega} + V_{2Ix} + V_{6\Omega} = 0$

$-10 + 4i_1 - 2Ix + 6(i_1 - i_2) = 0$

$4i_1 - 2(i_1 - i_2) + 6(i_1 - i_2) = 10$

$8i_1 + (-4)i_2 = 10$ (1)

Loop 2:

$V_{12V} + V_{6\Omega} + V_{2\Omega} = 0$

$12 + 6(i_2 - i_1) + 2i_2 = 0$

$(-6)i_1 + 8i_2 = -12$ (2)

2× (1) + (2)

$16i_1 + (-8)i_2 = 20$

$(-6)i_1 + 8i_2 = -12$

$10i_1 = 8$

$\therefore i_1 = 0.8$ A (3)

(3) in (2):

$(-6)0.8 + 8i_2 = -12$

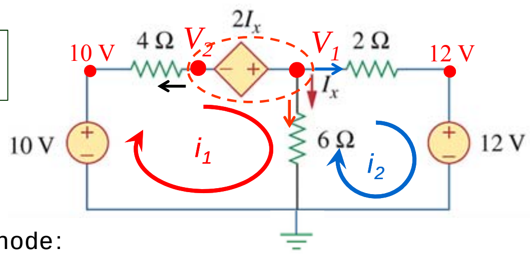
$\therefore i_2 = -0.9$ A

Problem 3-39 Determine i_1 and i_2 (Method#2 Supernode)

$$\boxed{i_1 = \frac{10 - V_2}{4}}$$

$$= \frac{3.2}{4}$$

$i_1 = 0.8\ A$



$$\boxed{i_2 = \frac{V_1 - 12}{2}}$$

$$= \frac{10.2 - 12}{2}$$

$i_2 = -0.9\ A$

KVL@Supernode:

$V_1 - V_2 = 2I_x$

$I_x = V_1/6$

$2V_1 - 3V_2 = 0$ (1)

KCL @ SN:

$i_{4\Omega} + i_{6\Omega} + i_{4\Omega} = 0$

x12{ $\frac{V_2 - 10}{4} + \frac{V_1}{6} + \frac{V_1 - 12}{2} = 0$

$3V_2 - 30 + 2V_1 + 6V_1 - 72 = 0$

$8V_1 + 3V_2 = 102$ (2)

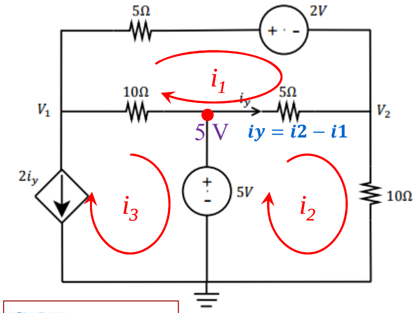
(1) + (2):

$(8+2)V_1 + (3-3)V_2 = 102$

$\therefore V_1 = 10.2\ V$

$\therefore V_2 = 6.8\ V$

2014: Use mesh analysis to find V_1, V_2 and i_y



$KVL@i3$

$i3 = -2iy = -2(i2 - i1)$

$2i1 - 2i2 - i3 = 0 \quad (1)$

$KVL@i1$

$(5 + 5 + 10)i1 - 5i2 - 10i3 + 2 = 0$

$20i1 - 5i2 - 10i3 = -2 \quad (2)$

$KVL@i2$

$(5 + 10)i2 - 5i1 - 5 = 0$

$-5i1 + 15i2 = 5 \quad (3)$

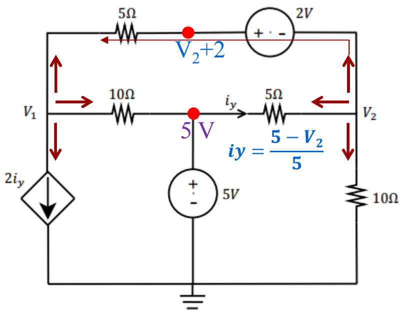
Solve
$i1 = -1.4\ A$
$i2 = -0.13\ A$
$i3 = -2.53\ A$

$iy = -0.13 - (-1.4)$
$= 1.27\ A$

$V1\ ?$
Using Ohms Law (V=RI) at 10Ω branch
$V1 - 5 = 10(i3 - i1)$
$V1 = 5 + 10\big(-2.53 - (-1.4)\big)$
$= 5 - 11.3 = -6.3\ V$

$V2\ ?$
Using Ohms Law (V=RI) at 5Ω branch
$5 - V2 = 5iy$
$V2 = 5 - 5(1.27)$
$= -1.35\ V$

2014: Use Nodal analysis to find V_1, V_2 and i_y



$KCL@V1$

$2iy + i_{10\Omega} + i_{5\Omega} = 0$

$$2i_y + \frac{V1 - 5}{10} + \frac{V1 - (V2 + 2)}{5} = 0$$

$$2\left(\frac{5 - V_2}{5}\right) + \frac{V1 - 5}{10} + \frac{V1 - (V2 + 2)}{5} = 0$$

$3V1 - 6V2 = -11 \quad (1)$

$KCL@V2$

$i_{10\Omega} + i_{5\Omega} + i_{2V} = 0$

$$\left(\frac{V2}{10}\right) + \frac{V2 - 5}{5} + \frac{(V2 + 2) - V1}{5} = 0$$

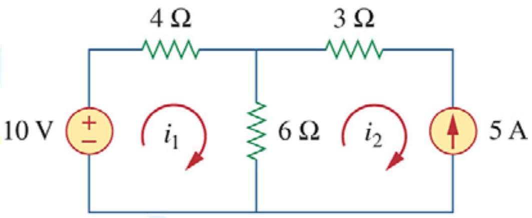
$-2V1 + 5V2 = 6 \quad (2)$

Solve
$V1 = -6.33\ A$
$V2 = -1.33\ A$

$$iy = \frac{5 - (-1.33)}{5}$$
$= 1.27\ A$

L#4: 3.5 Mesh Analysis With Current Sources

CASE #1: A current source exists only in one mesh: That loop current is then known.

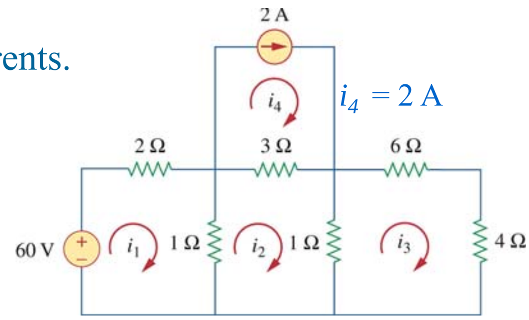


$i_2 = -5$ A

$-10 + 4i_1 + 6(i_1 - i_2) = 0$
$-10 + 10i_1 - 6(-5) = 0$
$10i_1 = -20$
$i_1 = -2$ A

Problem 3-46

Solve for the mesh currents.



$i_4 = 2$ A

KVL@Loop 1:

$-60 + 2i_1 + 1(i_1 - i_2) = 0$
$3i_1 - i_2 = 60 \qquad (1)$

KVL@Loop 2:

$1(i_2 - i_1) + 3(i_2 - 2) + 1(i_2 - i_3) = 0$
$-i_1 + 5i_2 - i_3 = 6 \qquad (2)$

KVL@Loop 3:

$1(i_3 - i_2) + 10i_3 = 0$
$-i_2 + 11i_3 = 0 \qquad (3)$

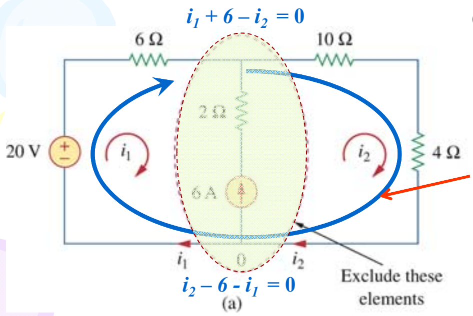
Solve for i_1, i_2 & i_3

Solve
$i1 = 21.9\ A$
$i2 = 5.68\ A$
$i3 = 0.52\ A$

3.5 Mesh Analysis
With Current Sources

CASE #2: A current source exists between two meshes: SUPERMESH

A *SUPERMESH* results when two meshes have a current source (dependent or independent), with or without other elements in series, in common.



Common source:
Note: i_2 flows in the same direction as the 6A source.

$$i_2 - i_1 = 6 \qquad i_1 = 2.8 - 6$$
$$i_1 = i_2 - 6 \quad (1) \qquad \therefore i_1 = -3.2 \text{ A}$$

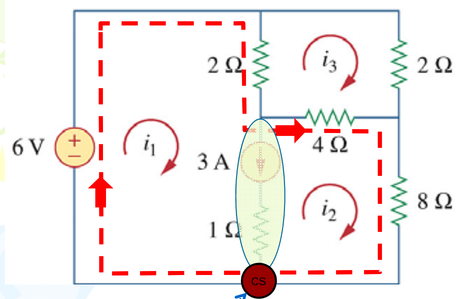
SUPERMESH:

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0 \quad (2)$$

(1) in (2)

$$-20 + 6(i_2 - 6) + 10i_2 + 4i_2 = 0$$
$$i_2 = 2.8 \text{ A}$$

PP 3-7 Determine i_1, i_2 and i_3.



Supermesh:

$$-6 + 2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2 = 0$$
$$2i_1 + 12i_2 - 6i_3 = 6 \quad (2)$$

Loop 3:

$$2(i_3 - i_1) + 2i_3 + 4(i_3 - i_2) = 0$$
$$-2i_1 - 4i_2 + 8i_3 = 0 \quad (3)$$

Note: 3A current source common to mesh i_1 & i_2

Common source: $i_1 - i_2 - 3 = 0$ (1)

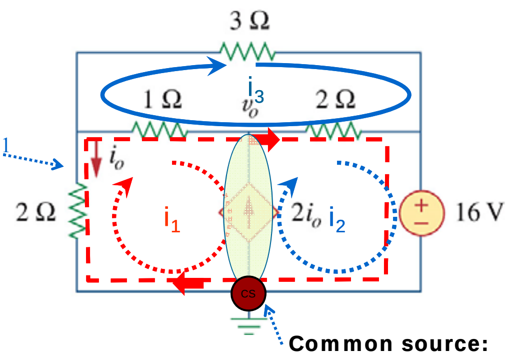
Solve for i_1, i_2 & i_3

Solve
$i1 = 3.47\ A$
$i2 = 0.47\ A$
$i3 = 1.11\ A$

Problem 3-46

Find v_0 and i_o

$i_o = - i_1$



Supermesh:

$$2i_1 + 1(i_1 - i_3) + 2(i_2 - i_3) + 16 = 0$$
$$3i_1 + 2i_2 - 3i_3 = -16 \quad (2)$$

Loop 3:

$$1(i_3 - i_1) + 3i_3 + 2(i_3 - i_2) = 0$$
$$-i_1 - 2i_2 + 6i_3 = 0 \quad (3)$$

Common source:

$$i_2 = 2i_o + i_1$$
$$i_2 = 2(-i_1) + i_1$$
$$i_1 + i_2 = 0 \quad (1)$$

Solve
$i_1 = -10.67$ A
$i_2 = 10.67$ A
$i_3 = 1.78$ A

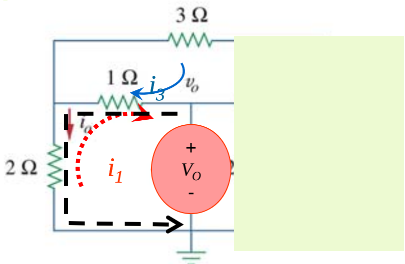
$i_o = 10.67$ A

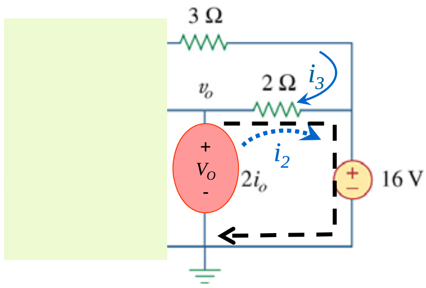
Problem 3-46

Find v_0

$i_1 = -10.67$ A
$i_2 = 10.67$ A
$i_3 = 1.78$ A

$i_o = 10.67$ A





KVL:

$$2i_1 + 1(i_1 - i_3) + v_o = 0$$
$$v_o = -3i_1 + i_3$$
$$v_o = 33.78\ V$$

KVL:

$$-v_o + 2(i_2 - i_3) + 16 = 0$$
$$v_o = 2(10.67 - 1.78) + 16$$
$$v_o = 33.78\ V$$

L#5: 3.6 Nodal and Mesh Analysis by inspection

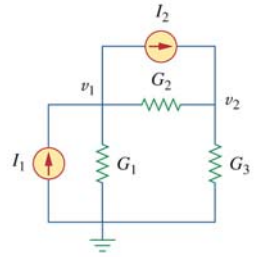
Special case #1: All sources = independent **current** sources

KCL@ V1 $G = \frac{1}{R}$

$I_1 - I_2 - i_{G1} - i_{G2} = 0$

$G_1 V_1 + G_2(V_1 - V_2) = I_1 - I_2$

$(G_1 + G_2) + (-G_2)V_2 = I_1 - I_2$



KCL@ V2

$I_2 = i_{G3} + i_{G2}$

$G_3 V_2 + G_2(V_2 - V_1) = I_2$

$(-G_2)V_1 + (G_2 + G_3)V_2 = I_2$

Matrix

$$\begin{array}{c} \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{cc} 1 & 2 \end{array} \\ \begin{bmatrix} G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 \end{bmatrix} \end{array} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} I_1 - I_2 \\ I_2 \end{bmatrix}$$

$$\begin{bmatrix} G_{11} & -G_{12} \\ -G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} I_{in} - Iout \\ I_{in} - Iout \end{bmatrix}$$

Note:

$G_{12} = G_{21}$

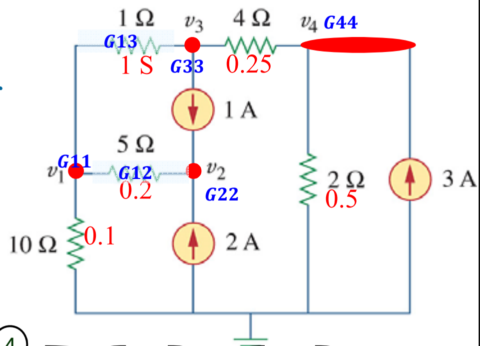
$G_{13} = G_{31}$

$G_{23} = G_{32}$

Conductance Matrix

- Diagonal elements (G_{11}, G_{22}, …) = *sum of all G's connected to that node.*
- Off-diagonal elements (G_{12}, G_{21}, …) = **–G** *connected between the two nodes.*

PP 3-8 By inspection, obtain the node voltage equations for the circuit in matrix form.



Current only source , use $\bar{G}.\bar{V} = \bar{I}$

Convert all R's into G's

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} (1+0.2+0.1) & -0.2 & -1 & 0 \\ -0.2 & 0.2 & 0 & 0 \\ -1 & 0 & (1+0.25) & -0.25 \\ 0 & 0 & -0.25 & (0.25+0.5) \end{bmatrix} \end{array} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2+1 \\ -1 \\ 3 \end{bmatrix}$$

Note:

$G_{12} = G_{21}$

$G_{13} = G_{31}$

$G_{14} = G_{41}$

$G_{23} = G_{32}$

$G_{24} = G_{42}$

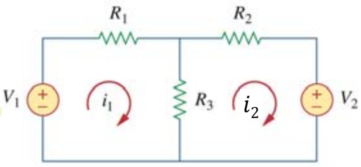
$G_{34} = G_{43}$

$$\begin{bmatrix} 1.3 & -0.2 & -1 & 0 \\ -0.2 & 0.2 & 0 & 0 \\ -1 & 0 & 1.25 & -0.25 \\ 0 & 0 & -0.25 & 0.75 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ -1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} G11 & -G12 & -G13 & -G14 \\ -G21 & G22 & -G23 & -G24 \\ -G31 & -G32 & G33 & -G34 \\ -G41 & -G42 & -G43 & G44 \end{bmatrix} \begin{bmatrix} v1 \\ v2 \\ v3 \\ v4 \end{bmatrix} = \begin{bmatrix} Iin - Iout \\ Iin - Iout \\ Iin - Iout \\ Iin - Iout \end{bmatrix}$$

3.6 Nodal and Mesh Analysis by inspection

Special case #2: All sources = independent **voltage** sources



KVL@ i_1

$$-V_1 + R_1 i_1 + R_3(i_1 - i_2) = 0$$

$$(R_1 + R_3)i_1 + (-R_3)i_2 = V_1$$

KVL@ i_2

$$V_2 + R_2 i_2 + R_3(i_2 - i_1) = 0$$

$$(-R_3)i_1 + (R_2 + R_3)i_2 = -V_2$$

$$\begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$$

Note:
$R_{12} = R_{21}$
$R_{13} = R_{31}$
$R_{14} = R_{41}$
$R_{23} = R_{32}$
$R_{24} = R_{42}$
$R_{34} = R_{43}$

$$\begin{array}{c c} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{bmatrix} R_{11} & -R_{12} \\ -R_{21} & R_{22} \end{bmatrix} \end{array}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} \Sigma V \\ \Sigma V \end{bmatrix}$$

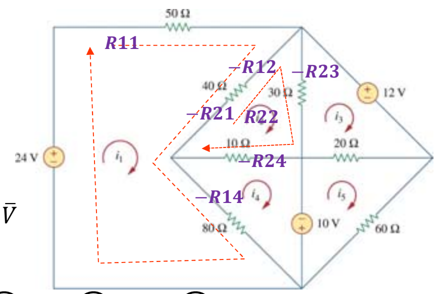
Resistance Matrix

- Diagonal elements (R_{11}, R_{22}, …) = *sum of all R's in the loop.*
- Off-diagonal elements (R_{12}, R_{21}, …) = *–R common to two loop between the two MESH*

Sum of all the voltage increases in the loop:
In @ – = +
In @ + = –
Remember: V is taken to the other side in the KVL equation.

PP 3-9 By inspection, obtain the mesh-current equations for the circuit in matrix form.

ONLY voltage source , use $\bar{R}.\bar{I} = \bar{V}$



Note:
$R_{12} = R_{21}$
$R_{13} = R_{31}$
$R_{14} = R_{41}$
$R_{23} = R_{32}$
$R_{24} = R_{42}$
$R_{34} = R_{43}$

$$\begin{array}{c c} & \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} & \begin{bmatrix} (50+40+80) & -40 & 0 & -80 & 0 \\ -40 & (40+30+10) & -30 & -10 & 0 \\ 0 & -30 & 30+20 & 0 & -20 \\ -80 & -10 & 0 & (10+80) & 0 \\ 0 & 0 & -20 & 0 & (20+60) \end{bmatrix} \end{array} . \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} \mathbf{24} \\ \mathbf{0} \\ \mathbf{-12} \\ \mathbf{10} \\ \mathbf{-10} \end{bmatrix}$$

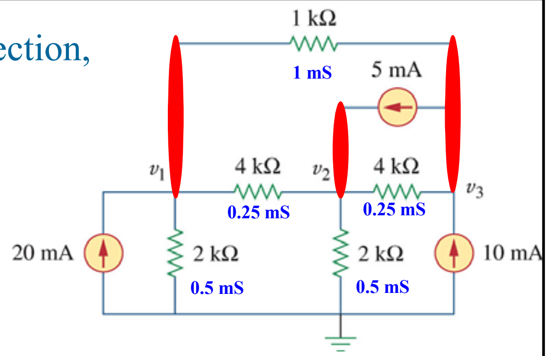
$$\begin{bmatrix} 170 & -40 & 0 & -80 & 0 \\ -40 & 80 & -30 & -10 & 0 \\ 0 & -30 & 50 & 0 & -20 \\ -80 & -10 & 0 & 90 & 0 \\ 0 & 0 & -20 & 0 & 80 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ -12 \\ 10 \\ -10 \end{bmatrix}$$

$$\begin{bmatrix} R11 & -R12 & -R13 & -R14 \\ -R21 & R22 & -R23 & -R24 \\ -R31 & -R32 & R33 & -R34 \\ -R41 & -R42 & -R43 & R44 \end{bmatrix}\begin{bmatrix} i1 \\ i2 \\ i3 \\ i4 \end{bmatrix} = \begin{bmatrix} \Sigma V \\ \Sigma V \\ \Sigma V \\ \Sigma V \end{bmatrix}$$

Problem 3-69 By inspection, obtain the node voltage equations for the circuit

Current only source , use $\bar{G}.\bar{V} = \bar{I}$

Convert all R's into G's



$$\begin{bmatrix} (0.25+0.5)m & -0.25m & -1m \\ -0.25m & (0.25+0.5+0.25)m & -0.25m \\ -1m & -0.25m & (0.25+1)m \end{bmatrix} . \begin{bmatrix} V1 \\ V2 \\ V3 \end{bmatrix} = \begin{bmatrix} 20m \\ 5m \\ (10-5)m \end{bmatrix}$$

X 1000

$$\begin{bmatrix} 0.75 & -0.25 & -1 \\ -0.25 & 1 & -0.25 \\ -1 & -0.25 & 1.25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 5 \\ 5 \end{bmatrix}$$

L#6:
3.7 Nodal versus Mesh Analysis

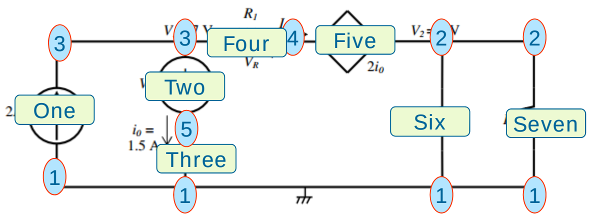
To select the method that results in the smaller number of equations. For example:

1. Choose nodal analysis for circuit with fewer nodes than meshes.
   *Choose mesh analysis for circuit with fewer meshes than nodes.
   *Networks that contain many series connected elements, voltage sources, or supermeshes are more suitable for mesh analysis.
   *Networks with parallel-connected elements, current sources, or supernodes are more suitable for nodal analysis.
2. If node voltages are required, it may be expedient to apply nodal analysis. If branch or mesh currents are required, it may be better to use mesh analysis.

Revision

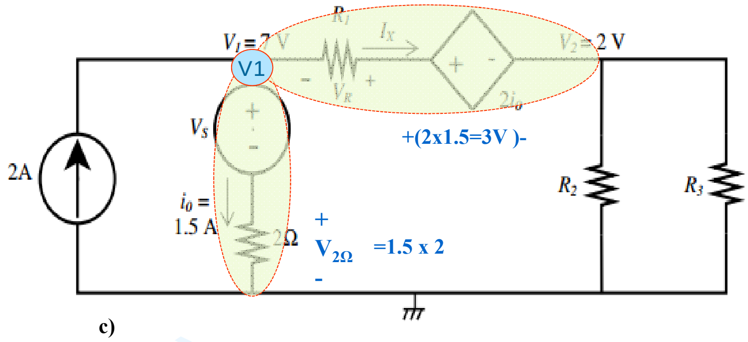
2012: Answer the questions

a) How many nodes are there in the circuit? [1]
b) How many branches are there in the circuit? [1]
c) Determine the voltage V_S. [1]
d) Determine the voltage V_R over resistor R_1. [2]
e) Determine the current I_X. [2]



2012: Source voltage V_S and resistors R_1, R_2 and R_3 are unknown:

c) Determine the voltage V_S. [1]
d) Determine the voltage V_R over resistor R_1. [2]
e) Determine the current I_X. [2]

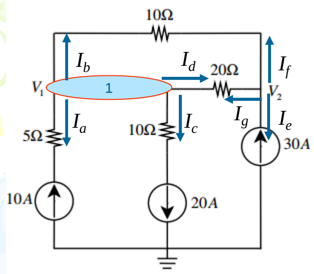


c)
$V_1 - 0 = V_S + V_{2\Omega}$
$7 = V_S + 3$
$V_S = 4$ V

d)
$V_1 - V_2 = -V_R + 2i_o$
$V_R = 2i_o - 7 + 2$
$= 3 - 7 + 2$
$V_R = -2$ V

e) KCL @ V1
$I_{in} = I_{out}$
$2 = I_X + 1.5$
$I_x = 0.5$ A

2012: Use node analysis to set up two equations of the following form for the circuit: $a\,V_1 + b\,V_2 = -200 \quad c\,V_1 + d\,V_2 = 600$

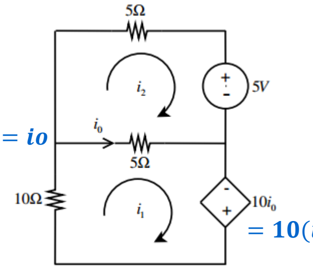


KCL@ V1
$I_a + I_b + I_C + I_d = 0$
$$-10 + \frac{V_1 - V_2}{10} + 20 + \frac{V_1 - V_2}{20} = 0$$
$$-200 + 2V_1 - 2V_2 + 400 + V_1 - V_2 = 0$$
$$3V_1 - 3V_2 = -200 \qquad (1)$$

KCL@ V2
$I_e + I_f + I_g = 0$
$$-30 + \frac{V_2 - V_1}{10} + \frac{V_2 - V_1}{20} = 0$$
$$-600 + 2V_2 - 2V_1 + V_2 - V_1 = 0$$
$$-3V_1 + 3V_2 = 600 \qquad (2)$$

2012: Use mesh analysis to set up two equations of the following form for the circuit: $a\,i_1 + b\,i_2 = 0 \quad c\,i_1 + d\,i_2 = -5$ **Solve for** i_o ***and power associated with dependable source***

$(i_1 - i_2) = io$



$= 10(i_1 - i_2)$

KVL@ i_2

$V_{5\Omega} + V_{5V} + V_{5\Omega} = 0$

$5i_2 + 5 + 5(i_2 - i_1) = 0$

$10i_2 + 5 - 5i_1 = 0$

$-5i_1 + 10i_2 = -5 \quad (2)$

$(1) + (2)$

$5i_1 + 5i_2 = 0$

$\underline{-5i_1 + 10i_2 = -5}$

$15i_2 = -5$

$i_2 = -0.33$ A

From (1)

$i_1 = -i_2 = 0.33\ A$

$i_o = i_1 - i_2 = 0.66\ A$

$P_d = -i_1 \times 10io$

$= -(0.33) \times 10 \times 0.66$

$= -2.178\ W$

KVL@ i_1

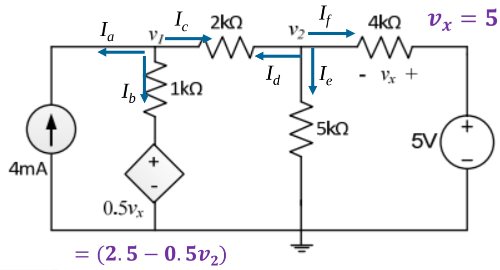
$V_{10\Omega} + V_{5\Omega} + V_{10io} = 0$

$10i_1 + 5(i_1 - i_2) - 10io = 0$

$10i_1 + 5(i_1 - i_2) - 10(i_1 - i_2) = 0$

$5i_1 + 5i_2 = 0 \quad (1)$

2014: Use nodal analysis to set up two equations of the following form: $a\,v_1 + b\,v_2 = 13 \qquad c\,v_1 - 19\,v_2 = d$



$v_x = 5 - v_2$

$= (2.5 - 0.5v_2)$

KCL@ v1

$I_a + I_b + I_C = 0$

$$-4m + \frac{v_1 - 0.5\,v_x}{1k} + \frac{v_1 - v_2}{2k} = 0$$

$-8 + 2v1 - (5 - v2) + v1 - v2 = 0$

$3\,v_1 - 0\,v_2 = 13 \quad (1)$

KCL@ v2

$I_d + I_e + I_f = 0$

$$\frac{v_2 - v_1}{2k} + \frac{v_2}{5k} + \frac{v_2 - 5}{4k} = 0$$

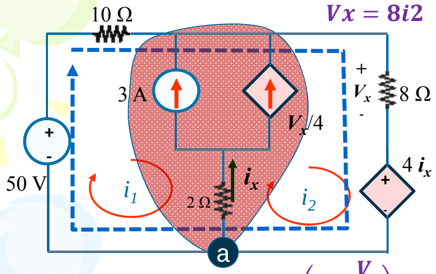
$10v2 - 10v1 + 4v2 + 5v2 - 25k = 0$

$-10v_1 + 19\,v_2 = 25k$

$10\,v_1 - 19\,v_2 = -25k \quad (2)$

Calculate i_x, and V_x

$Vx = 8i2$



$ix = \left(3 + \frac{V_x}{4}\right)$

KCL @ common node (a)

$i2 = i1 + ix$

$i2 = i1 + \left(3 + \frac{V_x}{4}\right)$

$i2 = i1 + 3 + \frac{8i2}{4}$

$-i1 - i2 = 3 \quad (1)$

KVL @ Supermesh

$-50 + 10i1 + 8i2 + 4ix = 0$

$10i1 + 16i2 = 38 \quad (2)$

$(2) + 10\,(1)$

$10i1 + 16i2 = 38$

$\underline{-10i1 - 10i2 = 30}$

$6i2 = 68$

$i2 = 11.38\ A$

$-i1 - (11.38) = 3 \quad (1)$

$i1 = -14.33\ A$

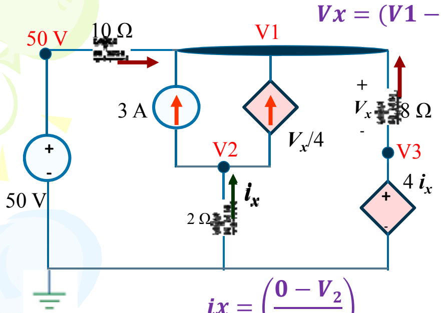
$V_x = 8i2 = 8 \times 11.38$

$= 90.67\ V$

$ix = \left(3 + \frac{90.67}{4}\right) = 25.67\ A$

Calculate i_x, and V_x (use Nodal Analysis)

$Vx = (V1 - V3)$



$ix = \left(\frac{0 - V_2}{2}\right)$

KCL @ V1

$i_{10\Omega} + 3 + \left(\frac{V_x}{4}\right) + i_{8\Omega} = 0$

$\frac{50 - V1}{10} + 3 + \frac{V1 - V3}{4} + \frac{V3 - V1}{8} = 0$

$2V1 - 10V3 = -640 \quad (1)$

KCL @ V2

$i_x = 3 + \left(\frac{V_x}{4}\right)$

$\frac{0 - V2}{2} = 3 + \frac{V1 - V3}{4}$

$-V1 - 2V2 + V3 = 12 \quad (2)$

KVL @ V3

$V3 = 4ix = 4\left(\frac{-V2}{2}\right)$

$2V2 + V3 = 0 \quad (3)$

Solve

V1= 193.33 V

V2=-51.33 V

V3=102.67 V

$ix = \left(\frac{-V2}{2}\right) = \left(\frac{-(-51.33}{2}\right)$

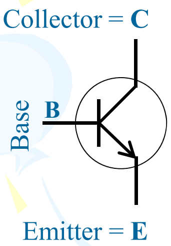
$= 25.67\ A$

$Vx = (V1 - V3) = 193.33 - 102.67$

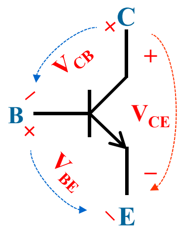
$= 90.67\ V$

L#7:
3.9 Transistor Circuits

BJT = Bipolar Junction Transistor



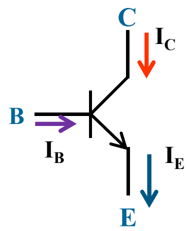
npn transistor



Voltage

KVL: $V_{CE} = V_{CB} + V_{BE}$

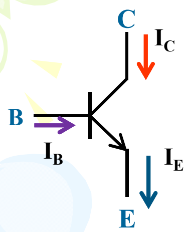
$V_{BE} = 0.7$ V



Currents

KCL: $I_E = I_C + I_B$

3.9 Transistor Circuits
Currents



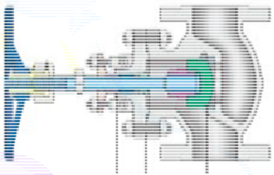
KCL: $I_E = I_C + I_B$

$\boxed{I_C = \beta I_B}$

$\therefore\ I_E = (1+\beta)I_B$

β = "Common-emitter current gain"

β ranges between 50 ⟶ 100



$\boxed{I_C = \alpha I_E}$

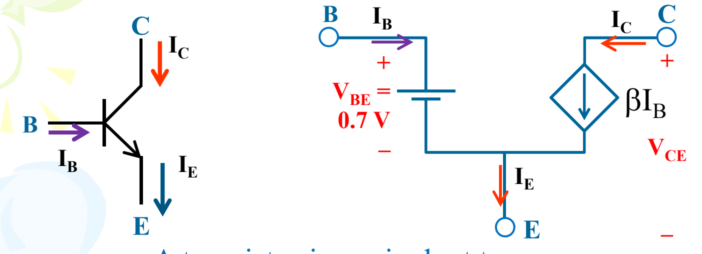
α = "Common-base current gain"

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}$$

α ranges between 0.98 ⟶ 0.999

3.9 Transistor Circuits

Equivalent network model



A transistor is equivalent to a current controlled current source.

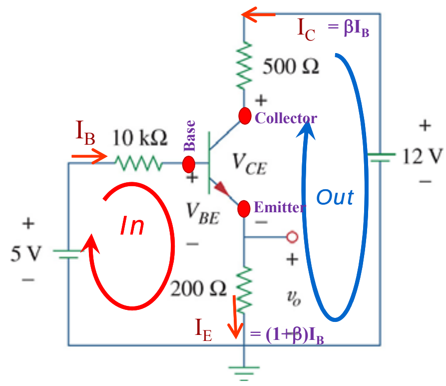
How does one analyse a circuit with a transistor in it?

Method #1: Mesh Analysis

Method #2: Node Analysis
Only applicable if the equivalent network model is used.

PP 3-12 Determine v_o and V_{CE} for the transistor circuit in which V_{BE} = 0.7 V and β = 100. **METHOD #1**

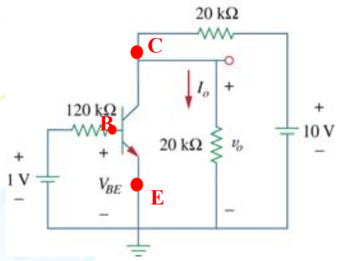
Identify Base, collector & Emitter
Indicate the currents: I_B, I_E and I_C
Use $I_E = (1+\beta)I_B$ and $I_C = \beta I_B$
Draw the input & output loops
$v_o = 200\ I_E$
$v_o = 200\ (1+\beta)I_B$ *(1)*
KVL at outer loop:
$12 - v_o - V_{CE} - 500 \times I_C = 0$
$12 - v_o - V_{CE} - 500 \times \beta I_B = 0$ *(2)*
Solve for I_B from Input loop.

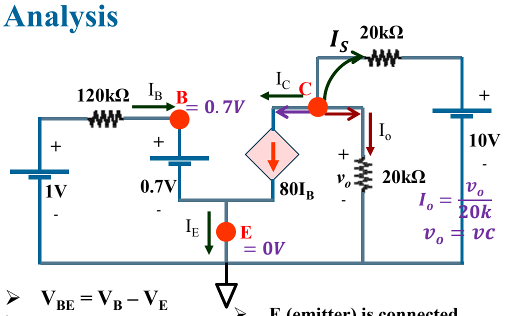


KVL for Input Loop:
$-5 + 10000 \times I_B + V_{BE} + v_o = 0$
$-5 + 10000 \cdot I_B + 0.7 + 200\ (1+\beta)I_B = 0$
$\therefore I_B = 0.1424$ mA = **142.4 μA**
$v_o = 200\ (100+1) \times 142.4\ \mu$ *(1)*
= **2.87 V**

$12 - v_o - V_{CE} - 500 \times \beta I_B = 0$ *(2)*
$12 - 2.87 - V_{CE} - 500 \times (100 \times 142.4\ \mu) = 0$
$V_{CE} = 12 - 2.87 - 7.12 =$ **2 V**

PP 3-13 Determine v_o and I_o for the transistor circuit in which $V_{BE} = 0.7$ V and $\beta = 80$. – **Node Analysis**





- $V_{BE} = V_B - V_E$
- $V_B = V_{BE} + V_E$
  $= 0.7 + 0$
- $V_B = 0.7$ V (3)
- E (emitter) is connected to GND @ 0 V

KCL @ Collector (C):

$I_C + I_o + I_S = 0$

$$80IB + \frac{v_o}{20000} + \frac{v_o - 10}{20000} = 0$$

$$20000 \times 80I_B + 2vo - 10 = 0$$

$$1600k\ IB + 2vo = 10 \quad (1)$$

$$I_B = \frac{1 - VB}{120k} \quad (2)$$

$$I_B = \frac{1 - 0.7}{120k} = 2.5\ \mu\text{A}$$

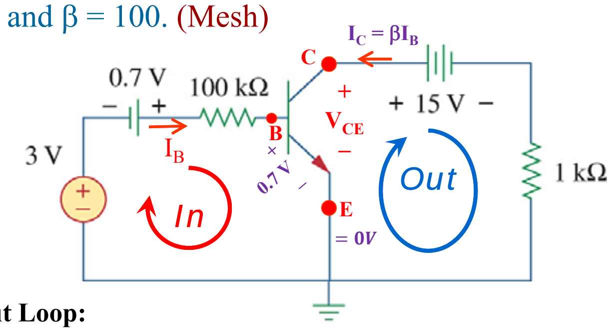
$$1600k \times 2.5\ \mu + 2vo = 10 \quad (1)$$

$$2vo = 10 - 4$$

$$v_o = 3\ \text{V}$$

$$I_o = \frac{v_o}{20k} = \frac{3}{20k} = 150\ \mu\text{A}$$

Problem 3-89 Determine V_{CE} and I_B for the transistor circuit in which $V_{BE} = 0.7$ V and $\beta = 100$. (Mesh)



KVL for Input Loop:

$-3 - 0.7 + 100k \cdot I_B + 0.7 = 0$

$I_B = 3 / 100k$

$= 30\ \mu$A

KVL for Output Loop:

$-V_{CE} + 15 + 1000 \cdot I_{out} = 0$

$15 + 1000 \cdot (-I_C) - V_{CE} = 0$

$V_{CE} = 15 + 1000 \cdot (-100 \cdot 30\mu)$

$V_{CE} = 15 - 1000 \cdot (100 \cdot 30\mu)$

$= 12$ V